

Probability HW4

Problem #1

$$a) \text{Var}(X_1) = E(X_1^2) - (E(X_1))^2 \quad E(X_1) = 0 \cdot \frac{3}{16} + 1 \cdot \frac{2}{16} + 2 \cdot \frac{5}{16} + 3 \cdot \frac{6}{16}$$

$$E(X_1^2) = 0^2 \cdot \frac{3}{16} + 1^2 \cdot \frac{2}{16} + 2^2 \cdot \frac{5}{16} + 3^2 \cdot \frac{6}{16} = 0 + \frac{2}{16} + 4 \cdot \frac{5}{16} + 9 \cdot \frac{6}{16}$$

$$= \frac{2}{16} + \frac{20}{16} + \frac{54}{16} = \frac{76}{16} = \frac{19}{4}$$

$$= \frac{15}{8}$$

$$\text{Var}(X_1) = \frac{19}{4} - \left(\frac{15}{8}\right)^2 = \frac{79}{64}$$

$$\text{Var}(X_2) = E(X_2^2) - (E(X_2))^2$$

$$E(X_2^2) = 1 \cdot \frac{1}{2} + 4 \cdot \frac{1}{2} = \frac{5}{2} \quad E(X_2) = \frac{3}{2}$$

$$\text{Var}(X_2) = \frac{5}{2} - \left(\frac{3}{2}\right)^2 = \frac{1}{4}$$

$$\text{Cov}(X_1, X_2) = E(X_1 X_2) - E(X_1)E(X_2)$$

$$E(X_1 X_2) = 0 \cdot 1 \cdot \frac{1}{8} + 0 \cdot 2 \cdot \frac{1}{16} + 1 \cdot 1 \cdot \frac{1}{16} + 1 \cdot 2 \cdot \frac{1}{16} + 2 \cdot 1 \cdot \frac{3}{8} + 2 \cdot 2 \cdot \frac{1}{8} + 3 \cdot 1 \cdot \frac{1}{8} + 3 \cdot 2 \cdot \frac{1}{4}$$

$$= \frac{1}{16} + \frac{2}{16} + \frac{1}{16} + \frac{2}{16} + \frac{6}{8} + \frac{4}{8} + \frac{3}{8} + \frac{6}{4}$$

$$= \frac{9}{16} + \frac{14}{16} + \frac{24}{16} = \frac{47}{16}$$

$$\text{Cov}(X_1, X_2) = \frac{47}{16} - \frac{15}{8} \cdot \frac{3}{2} = \frac{47}{16} - \frac{45}{16} = \frac{2}{16} = \frac{1}{8}$$

b) Conditional pmf of X_2 given $X_1=0$:

$$P(X_2=1 | X_1=0) = \frac{P(X_1=0, X_2=1)}{P(X_1=0)} = \frac{\frac{1}{8}}{\frac{3}{16}} = \frac{2}{3}$$

$$P(X_2=2 | X_1=0) = \frac{P(X_1=0, X_2=2)}{P(X_1=0)} = \frac{\frac{1}{16}}{\frac{3}{16}} = \frac{1}{3}$$

Problem #2

a) Based on the given equation, we know that

$$\int_0^{\infty} c e^{-2x} dx = 1 \Rightarrow c \int_0^{\infty} e^{-2x} dx = 1 \quad c \left[-\frac{1}{2} e^{-2x} \right]_0^{\infty} = 1$$

$$\Rightarrow c \left[0 + \frac{1}{2} \right] = 1 \Rightarrow c = 2$$

$$e^{-2x} \frac{d}{dx} = -2x \cdot e^{-2x} - \frac{1}{2x}$$

$$b) P(X > 2) = 1 - P(X \leq 2) = 1 - \int_0^2 2e^{-2x} dx = 1 - 2 \left[-\frac{1}{2} e^{-2x} \right]_0^2$$

$$= 1 - 2 \left[-\frac{e^{-4}}{2} + \frac{1}{2} \right] = 1 + e^{-4} - 1 = e^{-4}$$

Problem #3

$$a) f_x(x) = \int_0^{\infty} x e^{-(x+y)} dy = x e^{-x} \int_0^{\infty} e^{-y} dy = x e^{-x} [-e^{-y}]_0^{\infty}$$

$$= x e^{-x} [0 + 1] = x e^{-x}$$

$$b) f_y(y) = \int_0^{\infty} x e^{-x-y} dx = e^{-y} \int_0^{\infty} x e^{-x} dx = e^{-y} [-x e^{-x}]_0^{\infty} + \int_0^{\infty} e^{-x} dx$$

$$= e^{-y}$$

$$c) f_x(x) \cdot f_y(y) = x e^{-x} \cdot e^{-y} = x e^{-(x+y)} = f_{x,y}(x,y)$$

$\therefore X$ and Y are independent

Problem #4

$$E(X) = \frac{3}{5} \quad 1 = \int_0^1 f(x) dx = \int_0^1 a + bx^2 dx = [ax + \frac{1}{3}bx^3]'_0^1 = a + \frac{1}{3}b$$

$$\frac{3}{5} = \int_0^1 x \cdot f(x) dx = \int_0^1 ax + bx^3 dx = [\frac{1}{2}ax^2 + \frac{1}{4}bx^4]'_0^1 = \frac{a}{2} + \frac{1}{4}b$$

$$\begin{cases} a + \frac{1}{3}b = 1 \\ \frac{1}{2}a + \frac{1}{4}b = \frac{3}{5} \end{cases} \Rightarrow \begin{cases} b = \frac{6}{5} \\ a = \frac{3}{5} \end{cases}$$

Problem #5

$$a) f_x(x,y) = \int_0^{1-x} 3(x+y) dy = \int_0^{1-x} 3x + 3y dy = [3xy + \frac{3}{2}y^2]'_0^{1-x}$$

$$= 3x(1-x) + \frac{3}{2}(1-x)^2 = 3x - 3x^2 + \frac{3}{2} - 3x + \frac{3}{2}x^2$$

$$= -\frac{3}{2}x^2 + \frac{3}{2} = \frac{3}{2}(-x^2 + 1)$$

$$f_y(x,y) = \int_0^{1-y} 3(x+y) dx = \int_0^{1-y} 3x + 3y dx = [\frac{3}{2}x^2 + 3xy]'_0^{1-y}$$

$$= \frac{3}{2}(1-y)^2 + 3(1-y)y = \frac{3}{2} - 3y + \frac{3}{2}y^2 + 3y - 3y^2$$

$$= \frac{3}{2}(-y^2 + 1)$$

b)

$$E(X) = \int_0^1 x \cdot f_x(x,y) dx = \int_0^1 \frac{3}{2}(-x^3 + x) dx = \frac{3}{2} [-\frac{1}{4}x^4 + \frac{1}{2}x^2]'_0^1$$

$$= \frac{3}{2}(-\frac{1}{4} + \frac{1}{2}) = \frac{3}{2} \cdot \frac{1}{4} = \frac{3}{8}$$

$$E(X^2) = \int_0^1 x^2 f_x(x,y) dx = \int_0^1 \frac{3}{2}(-x^5 + x^3) dx = \frac{3}{2} [-\frac{1}{5}x^5 + \frac{1}{3}x^3]'_0^1$$

$$= \frac{3}{2}(-\frac{1}{5} + \frac{1}{3}) = \frac{3}{2} \cdot \frac{2}{15} = \frac{1}{5}$$

$$\text{Var}(X) = E(X^2) - (E(X))^2$$

$$= \frac{1}{5} - \left(\frac{3}{8}\right)^2 = \frac{1}{5} - \frac{9}{64} = \frac{19}{320}$$

c) $\text{Var}(X) = \text{Var}(Y) = \frac{19}{320}$ $E(X) = E(Y) = \frac{3}{8}$

$$E(X, Y) = 3 \int_0^1 y \int_0^{1-y} x(x+y) dx dy \Rightarrow \int_0^{1-y} x(x+y) dx = \left[\frac{1}{3}x^3 + \frac{1}{2}yx^2 \right]_0^{1-y} = \frac{1}{3}(1-y)^3 + \frac{1}{2}y(1-y)^2$$

$$\Rightarrow \int_0^1 y(1-3y+3y^2-y^3) + \frac{3}{2}y^2(1-2y+y^2) dy = \int_0^1 \frac{1}{2}y^4 - \frac{3}{2}y^2 + y dy$$

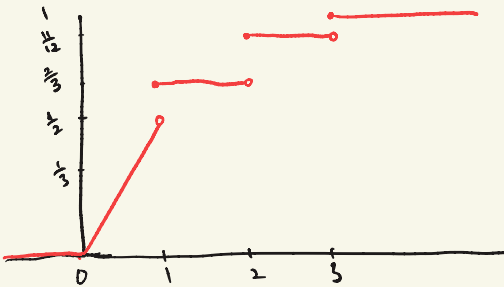
$$= \left[\frac{1}{10}y^5 - \frac{1}{2}y^3 + \frac{1}{2}y^2 \right]_0^1 = \frac{1}{10} - \frac{1}{2} + \frac{1}{2} = \frac{1}{10}$$

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)}\sqrt{\text{Var}(Y)}} = \frac{E(X, Y) - E(X)E(Y)}{\frac{19}{320}} = \frac{\frac{1}{10} - \left(\frac{3}{8}\right)^2}{\frac{19}{320}} = \frac{-\frac{13}{320}}{\frac{19}{320}} = -\frac{13}{19}$$

d) $f_{X|Y}(X|\frac{1}{2}) = \frac{f_{XY}(X, \frac{1}{2})}{f_Y(\frac{1}{2})} = \frac{3(X+\frac{1}{2})}{\frac{3}{2}(-\frac{1}{2})+1} = \frac{8}{3}(X+\frac{1}{2}) \quad \text{for } 0 \leq X \leq \frac{1}{2}$

Problem #6

a)



b) $P(X > 1/2) = 1 - P(X \leq 1/2) = 1 - F(1/2) = \frac{3}{4}$

c) $P(2 < X \leq 4) = P(X \leq 4) - P(X \leq 2) = F(4) - F(2) = 1 - \frac{1}{2} = \frac{1}{2}$

d) $P(X < 3) = F(3^-) = \frac{11}{12}$

e) $P(X=1) = F(1) - F(1^-) = \frac{2}{3} - \frac{1}{2} = \frac{1}{6}$